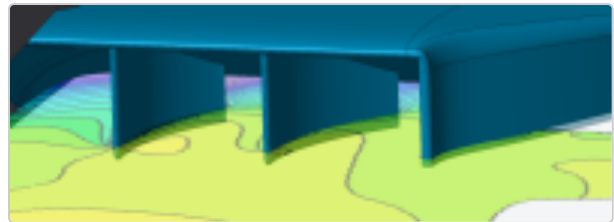
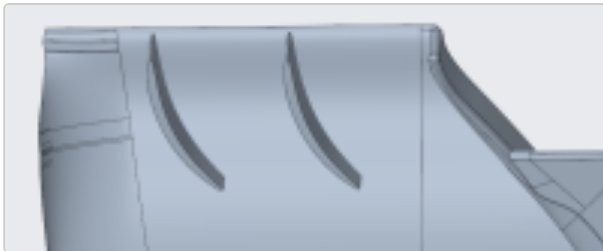


Vortexgenerators (VG)

Concept / Season Summary

Concept is same to last year: decreasing the pressure under the sidewing through a strong vortex core with low pressure inside.

This year we decided to screw them to the MW, so they can be changed after damage from cones. The VGs are airfoil shaped in cross-section to reduce energy loss and angle sensitivity in cornering. In addition they have a shedding edge on the bottom to help roll up a clean vortex with less losses. Other than that, only parameters were iterated to optimize AoA, length and position.



NOTE

Although a vortex can be used to reduce the boundary layer and help flow to stay attached, the main point of the VG in our case is to reduce the pressure under the MW. The vortex helps the flow to stay attached but only on one side where there is an upstream of air. Following the rotation it also pulls air down and away from the surface. This can cause separation especially in the trailing area of the VG where there is less energy in the air.